Activity 3

Determining Heart Rate

Materials :

Stop watch or Stop clock.

Procedure:

1. Have your partner to find the pulse in your wrist and count your heartbeats for 15 seconds while you are seated. Calculate your resting heart rate in beats per minute.
2. Have your partner to count your heart beats for 15 seconds after you jog or run for 5 minutes. Calculate your heart rate in beats per minute.

Analyse:

- What causes your pulse ?
- What causes the change in your heart beat rate in each situation ?

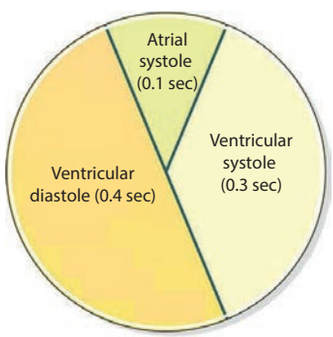
14.15.3 Cardiac Cycle

The sequence of events occurring from the **beginning** to the **completion of one heart beat** is called cardiac cycle. During cardiac cycle blood flows through the chambers of the heart in a specific direction. Each cardiac cycle lasts about **0.8 second**. The events during a single cardiac cycle involves

(a) Atrial systole: Contraction of auricles (0.1 sec)

(b) Ventricular systole: Contraction of ventricles (0.3 sec)

(c) Ventricular diastole: Relaxation of ventricles (0.4 sec)



14.15.4 Heart Sound

The rhythmic closure and opening of the valves cause the sound of the heart.

The **first sound LUBB** is of longer duration and is produced by the closure of the tricuspid and bicuspid valves after the beginning of ventricular systole. The **second sound DUPP** is of a shorter duration and produced by the closure of semilunar valves at the end of ventricular systole.

14.16 Blood pressure

Blood pressure is the **force exerted** during the flow of blood against the **lateral walls of arteries.** The blood pressure is high in the arteries gradually drops in the arterioles and capillaries and become very low in the veins.

Blood pressure is usually expressed in terms of the systolic pressure and diastolic pressure**.**

Systolic pressure: During ventricular systole, the left ventricle contracts and forces blood into the aorta. The pressure rises to a peak which is referred as systolic pressure.

Diastolic pressure: During diastole, the ventricles relax and the pressure falls to the lowest value which is referred as diastolic pressure**.**

In an healthy adult during normal resting condition systolic and diastolic blood pressure is expressed as **120mm / 80mm Hg.** Blood pressure varies during conditions of physical exercise, anxiety, emotions, stress and sleep.

A prolonged or constant elevation of blood pressure is a condition known as **hypertension (High blood pressure)** can increase the risk of heart attack and stroke. Decrease in blood pressure is termed hypotension **(Low blood pressure)**.

Stethoscope

A stethoscope is used to detect the sound produced by the internal organs of human body. The heart sound is heard by placing the

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